

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELGAVI-590018**



A MAJOR PROJECT REPORT ON

**“AI-Powered Real-Time Urban Traffic Congestion
Prediction Using Distributed Big Data Processing”**

Submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

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DON BOSCO INSTITUTE OF TECHNOLOGY

Kumbalagodu, Mysuru Road, Bengaluru-560074

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CERTIFICATE

This is to certify that the Major Project Report Entitled “**AI-Powered Real-Time Urban Traffic Congestion Prediction Using Distributed Big Data Processing**” is a Bonafede work carried out by **ADARSHA GC (1DB22CS003)**, **DINESH S (1DB22CS040)**, **SANKETH N (1DB21CS104)** & **SUDARSHAN (1DB23CS415)** in partial fulfilment for the award of the degree of Bachelor of Engineering in **Computer Science & Engineering** of Visvesvaraya Technological University, Belagavi, during academic year 2025-2026. All Corrections and Suggestions indicated in the Internal Assessment have been incorporated in the Major Project Report and it is deposited in the department library. The Major Project Report has been approved as it satisfies the academic requirements in respect of the Project Prescribed for the Bachelor of Engineering Degree in Computer Science & Engineering.

Signature of the Guide and HOD

Signature of the Principal

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EXAMINER

SIGNATURE WITH DATE

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DECLARATION

We, Adarsha GC, Dinesh S, Sanketh N, Sudarshan bearing USN: 1DB22CS003, 1DB22CS040, 1DB22CS193, 1DB23CS415 are the Students of 7th semester, B.E in Computer Science & Engineering, Don Bosco Institute of Technology, hereby declare that the entire work embodied in this Major Project Work titled, **“AI-Powered Real-Time Urban Traffic Congestion Prediction Using Distributed Big Data Processing”** has been carried out by us and submitted in partial fulfilment of the award of the degree in Bachelor of Computer Science & Engineering of the Visvesvaraya Technological University, Belagavi, during the academic year 2025-2026. Further the matter embodied in this report has not been submitted previously by anybody for the award of any degree equal to this or any other university.

Place: Bengaluru

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Date:

Place: Bengaluru

ABSTRACT

Urban traffic congestion has emerged as a critical challenge affecting mobility, fuel efficiency, and environmental sustainability in rapidly growing metropolitan areas. As cities expand, traditional traffic management systems—often reliant on static rules and historical averages—struggle to adapt to the stochastic and non-linear nature of modern vehicular flow. However, the advent of big data and the proliferation of real-time data streams from inductive loop sensors, GPS-enabled vehicles, and Internet of Things (IoT) devices have unlocked immense potential for the development of intelligent, adaptive traffic management systems.

This project investigates the application of advanced Deep Learning (DL) models for real-time traffic congestion prediction, specifically focusing on Recurrent Neural Network (RNN) architectures such as **Long Short-Term Memory (LSTM)** and **Gated Recurrent Units (GRU)**. Unlike traditional machine learning approaches (e.g., Support Vector Machines or ARIMA) that require extensive manual feature engineering, deep learning architectures enable automatic feature extraction from complex spatiotemporal traffic data. This capability allows the system to capture long-term temporal dependencies and subtle non-linear patterns, leading to significantly improved prediction accuracy, robustness, and scalability.

To handle the "Volume, Velocity, and Variety" of urban traffic data, this research also integrates a robust Big Data processing infrastructure. We review and implement a distributed pipeline utilizing **Apache Kafka** for fault-tolerant, real-time data ingestion and **Apache Spark** for high-speed stream processing. This architecture ensures that the deep learning models are fed with clean, synchronized, and normalized data with minimal latency.

The study presents a comprehensive evaluation of these methodologies using real-time data retrieved from the **TomTom API**. Experimental results demonstrate that the LSTM-based approach achieves superior performance, with a Coefficient of Determination (R^2) of **0.88**, effectively outperforming baseline models. Furthermore, the project culminates in the development of an interactive analytics dashboard that visualizes traffic heatmaps and predictive trends. By bridging the gap between theoretical deep learning models and practical big data deployment, this work provides a scalable blueprint for next-generation Intelligent Transportation Systems (ITS) within smart city infrastructures.

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